Detection of Suspicious Card Transactions Using Classical Machine Learning and Neural Network Approaches

Revnic Oana Cristina*

[1] Air Force Institute of Technology, Wright-Patterson AFB, OH 45433, USA
*Corresponding Author

my_gmail@gmail.com

Credit card fraud detection is a challenging machine learning problem because fraudulent transactions represent only a small fraction of all recorded activity while potentially causing significant financial losses. This study investigated fraud detection as a binary classification problem using a publicly available dataset containing 284,807 credit card transactions, including 492 fraudulent transactions (0.1727%) and 30 input features. The dataset was divided into training, validation, and holdout subsets, and class imbalance was addressed through class weighting.

Four modeling approaches were evaluated: a trivial majority-class classifier, Logistic Regression, Random Forest, and a multilayer perceptron (MLP) neural network. The best classical model was a Random Forest with class weights, which achieved Accuracy = 0.9994, Precision = 0.8621, Recall = 0.7576, F1-score = 0.8065, ROC-AUC = 0.9706, and PR-AUC = 0.7718 on the validation dataset. The best neural-network configuration, MLP (32,16) with L2 regularization ($\alpha = 0.01$), achieved F1-score = 0.1867 and PR-AUC = 0.6875. In comparison, the trivial classifier achieved Accuracy = 0.9983 but Precision = 0.0000, Recall = 0.0000, and F1-score = 0.0000 because it failed to identify any fraudulent transactions. On the holdout test set, the final Random Forest achieved Precision = 0.9302, Recall = 0.8163, F1-score = 0.8696, and PR-AUC = 0.8568, compared with F1-score = 0.2507 and PR-AUC = 0.7330 for the neural network.

The results demonstrate that accuracy alone is not an appropriate metric for highly imbalanced fraud-detection problems and that Random Forest substantially outperformed the neural-network and trivial baselines for this dataset.

Keywords: credit card fraud detection, transaction monitoring, imbalanced classification, supervised learning, classical machine learning, neural networks

1 Introduction & Background

Payment card fraud is a persistent financial oversight problem because fraudulent activity can be rare relative to legitimate transactions while still producing substantial financial and operational consequences. The European Banking Authority (EBA) and the European Central Bank (ECB) reported that the total value of fraud across major payment instruments in the European Economic Area amounted to €4.3 billion in 2022 and €2.0 billion during the first half of 2023 [1]. Although these totals include several payment instruments in addition to card payments, they demonstrate the continuing need for effective methods to identify suspicious activity within large volumes of otherwise legitimate transactions.

The problem is also relevant to public-sector financial oversight. Government organizations use purchase and travel card programs to support authorized expenditures, but these programs require safeguards capable of detecting illegal, improper, or erroneous purchases and payments. In a 2024 assessment of a federal purchase card program, the U.S. Agency for International Development Office of Inspector General stated that purchase-card risk assessments support efforts to prevent fraud, waste, and abuse in government-wide purchase card programs [2]. The present study does not use Department of Defense or other government transaction records. Instead, it uses a public, anonymized credit card transaction dataset as a proxy problem for examining analytical methods that could support suspicious-transaction screening in an organizational oversight setting.

Detecting fraudulent transactions is particularly challenging because fraud typically represents only a small minority of recorded activity. In a severely imbalanced dataset, a classifier may achieve high overall accuracy by predicting nearly all observations as legitimate while providing little operational value for identifying fraud. Prior research has therefore emphasized the need to account for class imbalance and to examine both traditional machine learning and deep learning approaches. Cherif et al. reviewed 40 credit card fraud detection studies published between 2015 and 2021 and identified class imbalance, feature engineering, traditional machine learning, and deep learning as central issues in the development of robust fraud detection systems [3].

This capstone project examines suspicious card transaction detection as a supervised binary classification problem using publicly available anonymized transaction data. The project compares the predictive performance of a classical machine learning approach with a neural network approach while accounting for the extreme imbalance of the fraudulent transaction class.

Research Question: Using publicly available anonymized card transaction data, how effectively can classical machine learning and neural network approaches identify fraudulent transactions in a highly imbalanced binary classification problem?

Table 1 *Modeling summary*

Attribute	This work
Problem type (Regression, Binary classification, Multi-class classification etc.)	Binary classification
Label description	Whether a card transaction is fraudulent or legitimate
Label variable name in the dataset	Class
Number of features in dataset	30 original input features
Number of datapoints	284,807 transactions
Hypothesis on which features will be most influential	Transaction amount and selected anonymized principal-component features are expected to contribute to distinguishing fraudulent from legitimate transactions. Because the principal-component features are anonymized, their original operational meanings cannot be interpreted.
Trivial model performance (validation)	Majority-class classifier: accuracy = 0.9983, fraud precision = 0.0000, fraud recall = 0.0000, fraud F1 = 0.0000, PR-AUC = 0.0017.
Best classical algorithm & performance (validation)	Random Forest with class_weight: accuracy = 0.9994, fraud precision = 0.8621, fraud recall = 0.7576, fraud F1 = 0.8065, PR-AUC = 0.7718.
Best NN & performance (validation)	MLP neural network (32, 16) with class weights and L2 regularization ($\alpha = 0.01$): accuracy = 0.9871, fraud precision = 0.1049, fraud recall = 0.8485, fraud F1 = 0.1867, PR-AUC = 0.6875.

The primary success criterion for the modeling phase was the ability to identify fraudulent transactions despite the severe imbalance of the output class. Because

fraudulent transactions represent only 0.1727% of the observations in the acquired dataset, overall accuracy alone would not provide an adequate measure of model usefulness [4]. A classifier that predicts nearly every transaction as legitimate could produce very high accuracy while failing to identify fraudulent activity. Therefore, the classical machine learning and neural network approaches were evaluated primarily using precision-recall based performance, together with precision, recall, F1-score, and confusion-matrix results. The final comparison also considered the ability of the selected models to generalize to previously unseen holdout data.

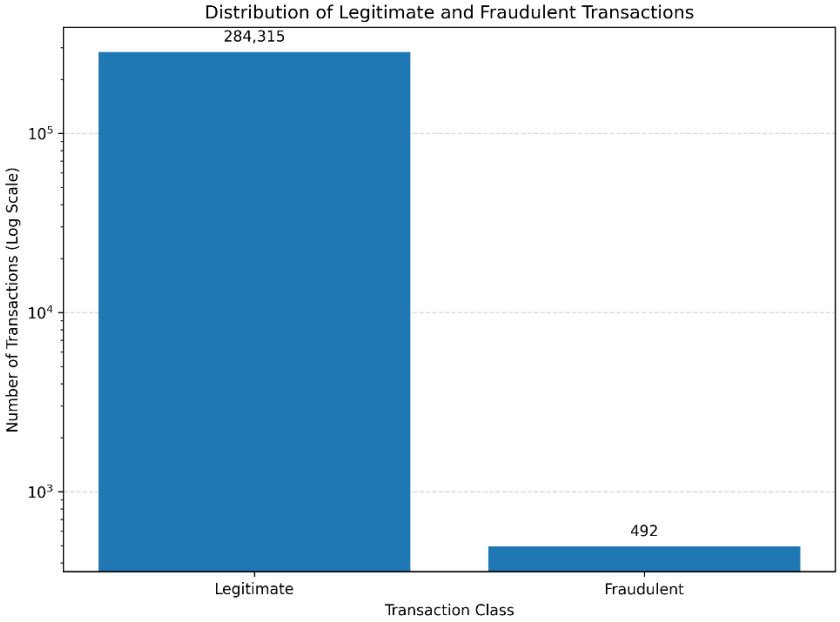


Figure 1 *Distribution of legitimate and fraudulent transactions in the acquired dataset. The logarithmic scale is used to display the substantially underrepresented fraudulent class [4].*

Table 2 *Prior machine learning analyses in credit card fraud detection*

Description of work	Method used	Performance	Ref
Compared multiple supervised machine learning models for credit card fraud detection using the Kaggle European cardholder dataset and SMOTE to address class imbalance.	Random Forest and XGBoost	Random Forest: Accuracy = 0.999 F1 = 0.872 AUC = 0.978 AUPRC = 0.871 XGBoost: Accuracy = 0.999 F1 = 0.837 AUC = 0.983	[5]

		AUPRC = 0.867	
Proposed a soft voting ensemble approach for imbalanced credit card fraud detection and compared it with sampling-based alternatives.	Soft voting ensemble using XGBoost, MLP, and KNN	Accuracy = 0.9996 F1 = 0.8764, AUROC = 0.9936	[6]
Compared supervised machine learning and artificial neural network approaches for credit card fraud detection.	Artificial Neural Network	Accuracy = 98.40% AUC = 92.88% Precision = 97.34%	[7]

These prior studies indicate that credit card fraud detection is commonly evaluated using metrics beyond accuracy, including F1-score, AUC, and AUPRC. The reported results also show that ensemble models and neural-network-based methods are frequently used for imbalanced fraud detection tasks, which supports the planned comparison between a classical machine learning approach and a neural network approach in this capstone project.

1.1 Data Acquisition

The selection of the dataset was motivated by recent work on credit fraud detection systems and their implementation in software and cloud environments. Teodoras et al. presented a Java microservice implementation for credit fraud detection using machine learning, while Popovici et al. described the DeFraudify4ALL prototype for fraud detection using big data and cloud technology [8], [9]. These studies motivated the selection of a transaction-level fraud detection problem that could support later comparison between a classical machine learning approach and a neural network approach.

The dataset used in this project was obtained from the publicly available Credit Card Fraud Detection dataset hosted on Kaggle by the Machine Learning Group of the Université Libre de Bruxelles [4]. The dataset contains anonymized credit card transactions made by European cardholders during a two-day period in September 2013. It was selected because it is based on real transaction data rather than purely simulated records, while still being publicly accessible for academic use. The dataset also has a high usability score on Kaggle and is widely used in credit card fraud detection research and demonstrations.

The downloaded file, dataset_creditcard.csv, contains 284,807 transaction records and 31 columns. The output variable is Class, where 0 represents a legitimate transaction and 1 represents a fraudulent transaction. The 30 original input features include Time, Amount, and 28 anonymized numerical variables labeled V1 through V28. The V1–V28 variables were transformed using principal component analysis to protect sensitive information, so their original operational meanings are not available. The Time variable records elapsed seconds from the first transaction rather than a calendar date, and Amount records the transaction value.

1.2 Data Understanding

The acquired dataset was examined to verify its structure, data quality, feature composition, and output-class distribution before any predictive modeling. The file contains 284,807 observations and 31 columns, consisting of 30 original input features and one binary output variable. All variables are numerical: 30 columns are stored as floating-point variables and the target variable, Class, is stored as an integer variable. No missing values were identified in the dataset. The initial quality review identified 1,081 exact duplicate rows; these observations were documented but not removed during this descriptive phase because any decision to remove records should be justified during later model-development work.

The output variable is severely imbalanced. The dataset contains 284,315 legitimate transactions and 492 fraudulent transactions, meaning that fraudulent transactions represent only 0.1727% of all observations [4]. This imbalance is central to the modeling problem because a classifier could achieve high overall accuracy by predicting nearly all transactions as legitimate. Therefore, the modeling work required evaluation metrics that emphasize minority-class detection rather than accuracy alone.

The `Amount` variable shows different descriptive patterns across the two output classes. Legitimate transactions have a mean amount of 88.29 and a median amount of 22.00, while fraudulent transactions have a mean amount of 122.21 and a median amount of 9.25. This difference between the mean and median values indicates that transaction amounts are right-skewed, supporting the use of a logarithmic transformation for visualization. The `Time` variable spans approximately 48 elapsed hours and records seconds since the first transaction rather than a calendar date. For descriptive analysis, elapsed time was converted to elapsed hours to examine the timing of fraudulent observations across the available two-day period.

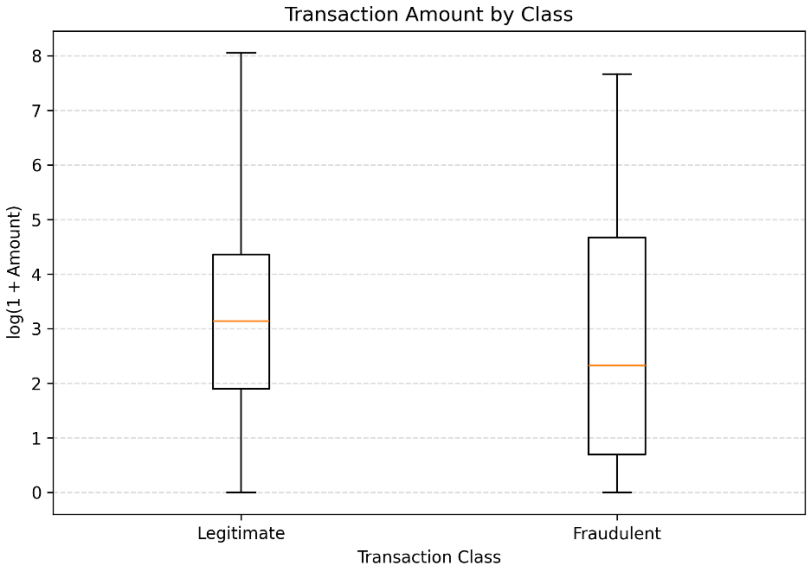


Figure 2 Comparison of transaction amounts for legitimate and fraudulent observations. The transformation log(1 + Amount) is applied for visualization because transaction values are right-skewed. Figure generated by the author using the Credit Card Fraud Detection dataset [4].

Figure 2 compares transaction amounts after applying the log(1 + Amount) transformation for visualization. On this transformed scale, the median line for legitimate transactions appears around 3.1, corresponding to a raw median amount of 22.00, while the median line for fraudulent transactions appears around 2.3, corresponding to a raw median amount of 9.25. This indicates that fraudulent transactions have a lower median amount than legitimate transactions. However, the descriptive statistics show that the mean amount is higher for fraudulent transactions, 122.21 compared with 88.29 for legitimate transactions, suggesting that some larger fraudulent transactions increase the average. Therefore, transaction amount may contain useful signal, but it should not be used alone for fraud detection.

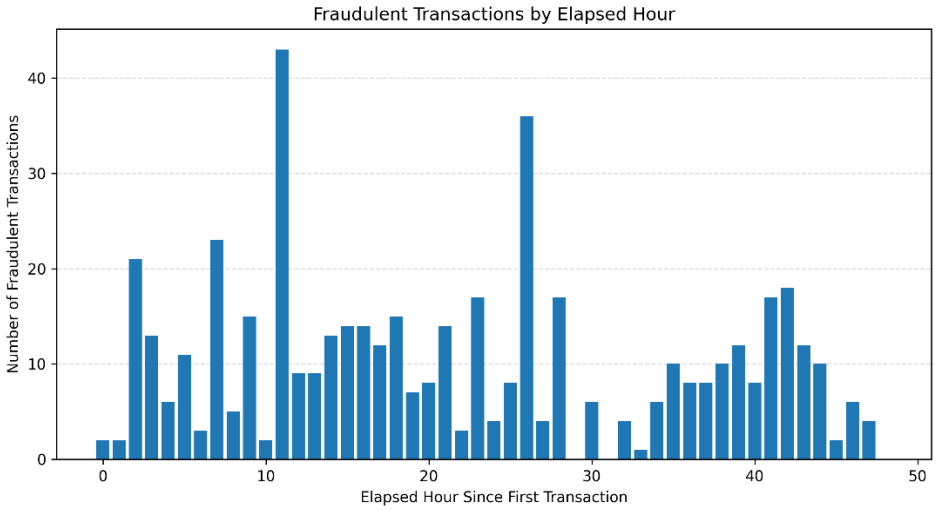


Figure 3 Distribution of fraudulent transactions across elapsed hours since the first recorded transaction. The elapsed-time variable does not represent calendar dates or clock time. Figure generated by the author using the Credit Card Fraud Detection dataset [4].

Figure 3 shows the number of fraudulent transactions in each elapsed-hour interval since the first recorded transaction. Fraudulent observations appear across most of the approximately 48-hour observation period, although some intervals contain noticeably higher counts. The largest concentrations occur around elapsed hour 11 and elapsed hour 26, suggesting that fraudulent activity is not uniformly distributed across time. However, the elapsed-time variable should be interpreted carefully because it does not represent a calendar date or clock time. In later modeling, time-based information should be evaluated in a way that avoids creating a predictor that may not generalize to future transaction data.

Table 3 Feature Summary

Input or Output Variable	Role	Data Understanding Result
Time	Input	Numerical variable; records seconds elapsed since the first transaction; spans approximately 48 hours
Amount	Input	Numerical variable; right-skewed transaction value; median = 22.00 for legitimate transactions and

		9.25 for fraudulent transactions
V1-V28	Input	Twenty-eight anonymized PCA-transformed numerical features; original operational meanings are not available
Class	Output	Binary target variable; 0 = legitimate and 1 = fraudulent
Observations	Dataset summary	284,807 total transactions
Class distribution	Output distribution	284,315 legitimate transactions and 492 fraudulent transactions; fraud = 0.1727%

Because this is a classification problem, the number and balance of output classes are especially important. The dataset contains two classes, legitimate and fraudulent, but the fraudulent class is substantially underrepresented. This confirms that the project is not only a binary classification task, but also an imbalanced classification problem. The imbalance influenced preprocessing, model selection, and performance evaluation [4].

2 Method

The development of the fraud-detection modeling framework followed the Cross-Industry Standard Process for Data Mining (CRISP-DM). This methodology provides a structured and iterative approach for data-driven projects and consists of six phases: business understanding, data understanding, data preparation, modeling, evaluation, and deployment.

During the business understanding phase, the objective was defined as identifying fraudulent credit card transactions in a highly imbalanced binary classification problem. The data understanding phase involved exploring the publicly available Credit Card Fraud Detection dataset, examining class distributions, reviewing descriptive statistics, and visualizing key variables. The data preparation phase included data-quality assessment, verification of missing values and duplicate observations, and preparation of the input variables for machine learning analysis.

The modeling phase used a stratified train-validation-test split to preserve the minority fraud class across all subsets. Multiple classification approaches were evaluated, including a trivial majority-class baseline model, classical machine learning algorithms, and a neural network model. Because fraudulent transactions represented only 0.1727% of the observations, class-weighting techniques were incorporated to reduce the impact of class imbalance during model training.

The evaluation phase focused on metrics appropriate for imbalanced classification problems, including precision, recall, F1-score, ROC-AUC, PR-AUC, and confusion matrices. Validation-set performance was used to compare candidate models while the holdout test set was reserved for the final evaluation of the selected models.

2.1 Data Preparation

The publicly available Credit Card Fraud Detection dataset was reviewed prior to model development. A data-quality assessment identified no missing values across the 31 variables. The dataset contained 1,081 duplicate observations. Because duplicate transactions may represent legitimate repeated activity and have been retained in prior studies using this dataset, they were not removed for the current analysis.

The input variables consisted of the elapsed-time variable (Time), transaction amount (Amount), and 28 anonymized principal-component variables (V1–V28). The output variable was Class, where 0 represents a legitimate transaction and 1 represents a fraudulent transaction.

Several preparation steps were performed before modeling. The dataset was partitioned into training, validation, and test subsets using a stratified 60/20/20 split. Stratification was used to preserve the extremely rare fraud class across all subsets. The training set was used to fit the models, the validation set was used for model comparison and selection, and the holdout test set was reserved for the final evaluation.

Feature scaling was applied through a modeling pipeline using RobustScaler. This approach reduces the influence of extreme values while preventing information leakage because scaling parameters are estimated only from the training data. To address the severe class imbalance, class-weighting techniques were incorporated into the machine learning and neural-network models. No oversampling or synthetic-sample generation techniques were applied during this phase of the project.

The class distribution shown in Figure 1 confirms the severe imbalance of the dataset, with fraudulent transactions representing only 0.1727% of all observations. This imbalance motivated the use of stratified data partitioning and class-weighting during model development.

2.2 Metrics

Because this project is a highly imbalanced binary classification problem, model performance was not evaluated using accuracy alone. Fraudulent transactions represent only 0.1727% of the dataset, so a model can achieve high accuracy while failing to identify any fraudulent transactions. This issue was demonstrated by the trivial majority-class classifier, which achieved 0.9983 validation accuracy but produced zero true positives, resulting in fraud precision, recall, and F1-score values of 0.0000.

The primary model comparison metrics were fraud-class precision, fraud-class recall, fraud-class F1-score, and precision-recall area under the curve (PR-AUC). Precision measures the proportion of transactions predicted as fraudulent that were actually fraudulent, while recall measures the proportion of actual fraudulent transactions detected by the model. F1-score was used as a balanced summary of precision and recall.

PR-AUC was emphasized because it is more informative than accuracy or ROC-AUC when the positive class is rare.

ROC-AUC and accuracy were also reported as secondary metrics. ROC-AUC provides a threshold-independent measure of ranking performance, while accuracy is included only to support comparison with the trivial baseline. Confusion matrices were used to report true positives, false positives, false negatives, and true negatives, allowing the operational tradeoff between missed fraud and false alerts to be examined.

2.3 Classical Modeling

Two classical machine learning models were evaluated during this iteration: Logistic Regression and Random Forest. Logistic Regression was selected as an interpretable linear baseline for binary classification, while Random Forest was selected as a nonlinear ensemble method capable of modeling interactions among the input variables. These models provide a useful contrast between a linear classifier and a tree-based ensemble classifier, both of which are commonly used in credit card fraud detection research [10].

The severe class imbalance shown in Fig. 1 was addressed using class-weighting. For Logistic Regression, class_weight='balanced' was used to assign larger penalties to errors on the minority fraud class. For Random Forest, class_weight='balanced' was also used so that the tree-splitting process accounted for the unequal class frequencies. The balanced weighting approach assigns weights inversely proportional to class frequencies, reducing the tendency of the algorithms to predict only the majority class [11].

Overfitting was monitored through the train-validation-test structure described in Section 2.1. Models were fitted only on the training set, compared using the validation set, and the test set was reserved for final evaluation. This structure reduces the risk of selecting a model based on holdout test performance. The Random Forest model was constrained using a maximum tree depth of 12 and 100 trees to reduce excessive model complexity while still allowing nonlinear relationships among the variables. For the final holdout evaluation, the Random Forest was refitted on the training set with 300 trees and no depth limit, using the same class weighting.

Input-variable contribution was assessed for the Random Forest model using feature importance values. As shown in Fig. 4, the most influential features were V14, V10, V12, V11, and V4. Because most predictors are anonymized PCA-transformed components, these values indicate relative predictive contribution but do not provide direct operational interpretation.

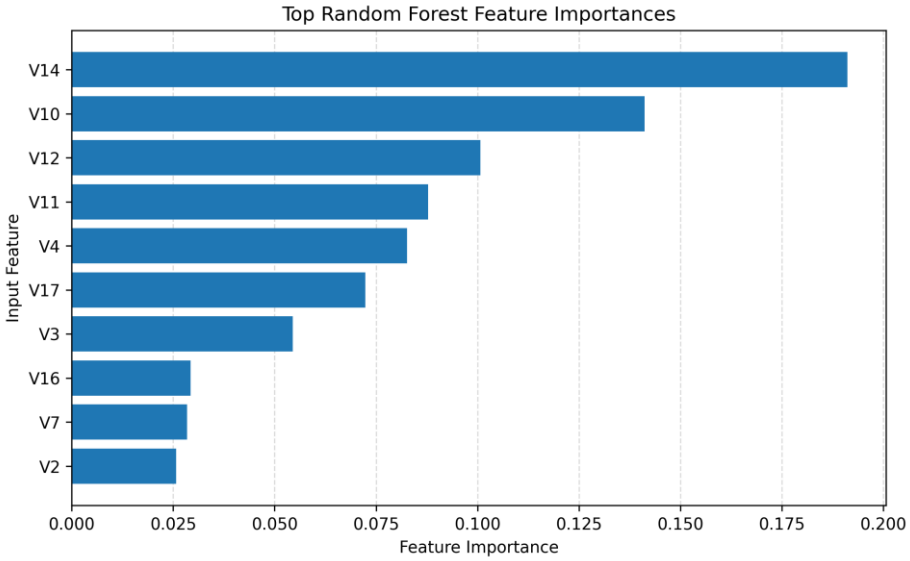


Figure 4 *Top ten Random Forest feature importance values. Figure generated by the author using the Credit Card Fraud Detection dataset [4].*

Figure 4 shows that V14, V10, V12, V11, and V4 were the most influential predictors in the Random Forest model. V14 had the highest importance value (0.1911), followed by V10 (0.1412) and V12 (0.1007). Because the predictors are anonymized PCA-transformed variables, these results indicate relative model contribution rather than the operational meaning of specific transaction characteristics.

2.4 Neural Network Modeling

A multilayer perceptron (MLP) neural network was evaluated as the neural-network approach for the binary fraud detection task. The MLP was selected because the dataset consists of fixed-length numerical input features rather than images, text, or sequential records requiring a specialized neural architecture. The network used ReLU activation functions in the hidden layers and a binary output structure suitable for distinguishing legitimate transactions from fraudulent transactions.

The neural network was trained using the Adam optimizer [12]. Adam was selected because it is an adaptive stochastic gradient-based optimizer commonly used for neural network training. Because this task is binary classification, the model was optimized using a binary classification objective. In the scikit-learn implementation used for this iteration, MLPClassifier uses a logistic output function for binary classification and optimizes log-loss [13].

To reduce overfitting, the neural network was trained within the same train-validation-test framework used for the classical models. The model was fitted on the training set, compared on the validation set, and the test set was reserved for final evaluation. Early stopping was enabled with a 10% internal validation fraction from the training

data, allowing training to stop when validation performance no longer improved. The input variables were scaled using RobustScaler inside the modeling pipeline so that scaling parameters were estimated from the training data only.

Because the fraudulent class was severely underrepresented, class weights were incorporated through sample weights during neural-network training. A small hyperparameter sweep compared three candidate hidden-layer architectures: (8), (16, 8), and (32, 16). The best neural network on the validation set was the MLP with hidden layers (32, 16), which achieved accuracy = 0.9871, fraud precision = 0.1047, fraud recall = 0.8485, fraud F1 = 0.1865, and PR-AUC = 0.6867.

3 Analysis & Results

This section summarizes the validation results for the trivial baseline, classical machine learning models, and neural network models. Model comparison was performed on the validation set to avoid using the holdout test set during model selection. Holdout test results for the final models are reported in Section 3.3.

3.1 Classical Modeling

The classical modeling analysis compared a trivial majority-class classifier with two class-weighted machine learning models: Logistic Regression and Random Forest. Model comparison was performed on the validation dataset after fitting each model on the training dataset. Because fraudulent transactions represent only 0.1727% of the observations, a trivial baseline was included to demonstrate the limitations of accuracy as a standalone performance metric.

Table 4 *Classical modeling validation results*

Model	ROC-AUC	PR-AUC	F1	Precision	Recall	Accuracy
Random Forest with class_weight	0.9706	0.7718	0.8065	0.8621	0.7576	0.9994
Logistic Regression with class_weight	0.9747	0.6794	0.1116	0.0595	0.8990	0.9751
Trivial majority-class classifier	0.5000	0.0017	0.0000	0.0000	0.0000	0.9983

Table 4 summarizes validation-set performance for the trivial baseline and the classical machine learning models. The Random Forest model achieved the strongest balanced performance, with F1 = 0.8065, precision = 0.8621, recall = 0.7576, and PR-AUC = 0.7718. Although Logistic Regression achieved higher recall, 0.8990, it produced much lower precision, 0.0595, indicating a substantially larger number of false positives. The trivial majority-class classifier achieved high accuracy, 0.9983, but failed

to identify any fraudulent transactions, producing precision, recall, and F1 values of 0.0000.

Random Forest was selected as the best classical model because it provided the strongest balance between fraud detection and false-positive control. It achieved the highest F1-score and PR-AUC among the classical models, while maintaining high precision. This balance is more useful for fraud screening than accuracy alone, because excessive false positives can increase investigation workload.

3.2 Neural Network Modeling

Three neural-network architectures were evaluated using the validation dataset. The baseline architecture contained a single hidden layer with eight neurons. Additional experiments evaluated two-hidden-layer architectures containing (16, 8) and (32, 16) neurons. The objective of the hyperparameter sweep was to determine whether increasing model capacity improved fraud-detection performance [14].

Table 5 *Neural network modeling validation results.*

Model	ROC-AUC	PR-AUC	F1	Precision	Recall	Accuracy
Baseline NN: MLP (8,)	0.9446	0.6874	0.0939	0.0496	0.8687	0.9709
After hyperparameter sweep: MLP (32,16)	0.9555	0.6867	0.1865	0.1047	0.8485	0.9871
After regularization: MLP (32,16), α=0.01	0.9555	0.6875	0.1867	0.1049	0.8485	0.9871

Table 5 summarizes validation-set performance for the neural-network models. The baseline neural network, MLP (8,), achieved F1 = 0.0939 and PR-AUC = 0.6874. Increasing the network size improved performance substantially. The MLP (32, 16) architecture achieved F1 = 0.1865, approximately double the baseline model, while also improving ROC-AUC to 0.9555.

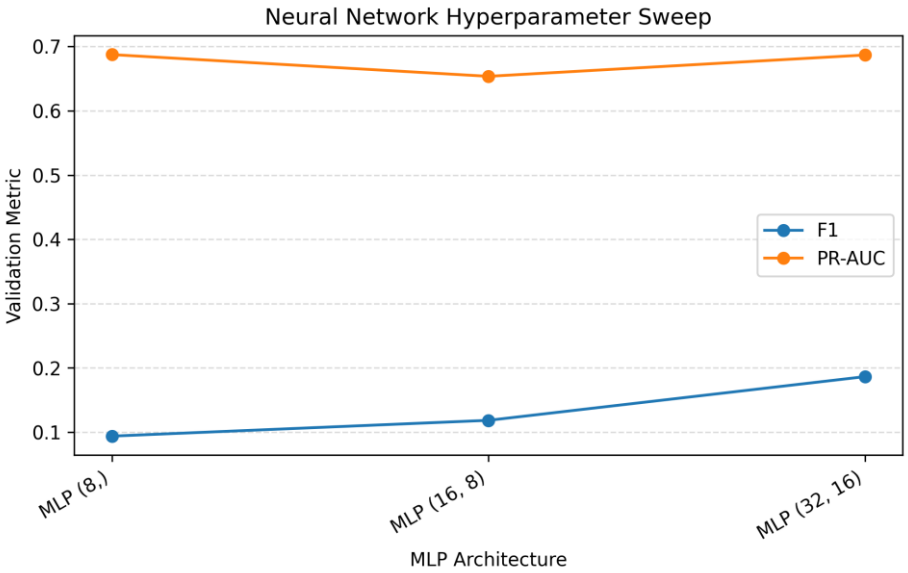


Figure 5 *Validation-set F1-score and PR-AUC for the neural-network architectures evaluated during the hyperparameter sweep.*

Figure 5 summarizes the effect of neural-network architecture on validation-set performance. Increasing model complexity improved fraud-detection performance. The MLP (32, 16) architecture achieved the highest F1-score, 0.1865, while maintaining a competitive PR-AUC value of 0.6867. The MLP (16, 8) architecture produced intermediate performance, and the smallest network, MLP (8,), achieved the lowest F1-score. The results suggest that increasing the number of hidden layers and neurons improved the model's ability to distinguish fraudulent from legitimate transactions. However, the differences in PR-AUC across architectures were relatively small, indicating that architecture size had a larger effect on F1-score than on ranking performance.

A regularization experiment was then performed using L2 regularization. The best result was obtained with alpha = 0.01. However, the resulting performance changes were minimal. The F1-score increased only from 0.1865 to 0.1867 and PR-AUC increased from 0.6867 to 0.6875. These results suggest that the selected neural-network architecture was not strongly affected by the tested regularization values.

The MLP (32, 16) architecture with alpha = 0.01 was selected as the best neural-network model because it achieved the highest F1-score among the evaluated neural-network configurations while maintaining comparable ROC-AUC and PR-AUC values. Although the improvement from regularization was small, this configuration produced the strongest overall neural-network performance on the validation dataset.

3.3 Model Evaluation

The final models were evaluated on the holdout test dataset using the metrics defined in Section 2.2. Because fraudulent transactions represent only 0.1727% of all observations, model evaluation focused primarily on Precision, Recall, F1-score, ROC-AUC, and PR-AUC rather than overall accuracy. Results from the final Random Forest and Neural Network models were compared against the project objectives and the trivial baseline classifier [15].

Table 6 *Final Results*

Model	ROC-AUC	PR-AUC	F1	Precision	Recall	Accuracy
Random Forest	0.9803	0.8568	0.8696	0.9302	0.8163	0.9996
Neural Network	0.9617	0.7330	0.2507	0.1457	0.8980	0.9908

The Random Forest model achieved the strongest overall performance on the holdout dataset. It obtained an F1-score of 0.8696, Precision of 0.9302, Recall of 0.8163, ROC-AUC of 0.9803, and PR-AUC of 0.8568. The Neural Network achieved higher Recall (0.8980), detecting 88 of the 98 fraudulent transactions in the holdout set compared with 80 for the Random Forest. However, this came at the cost of much lower Precision (0.1457), meaning that many more legitimate transactions were incorrectly flagged as fraudulent. Compared with the trivial majority-class classifier evaluated earlier in the project, both machine learning models provided substantial improvements. The trivial model achieved high accuracy because of the extreme class imbalance but failed to identify fraudulent transactions. This confirms that accuracy alone is not a sufficient metric for fraud-detection problems.

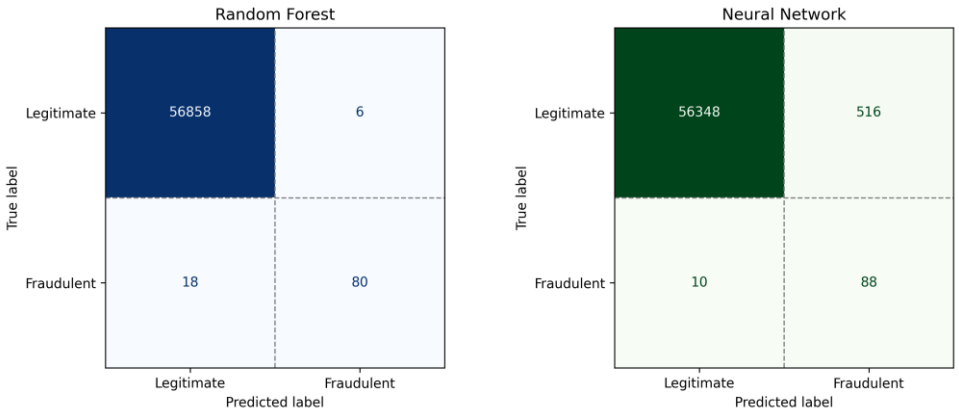


Figure 6 *Holdout confusion matrices for the final Random Forest and Neural Network models.*

Figure 6 shows the classification outcomes of the final models. The Random Forest correctly classified 56,858 legitimate transactions and 80 fraudulent transactions, missed 18 fraud cases, and generated only 6 false positives. The Neural Network correctly identified 88 fraudulent transactions and missed only 10 fraud cases, but produced 516 false positives. Although the Neural Network achieved higher Recall, the Random Forest produced 510 fewer false alerts and therefore achieved much higher Precision and F1-score.

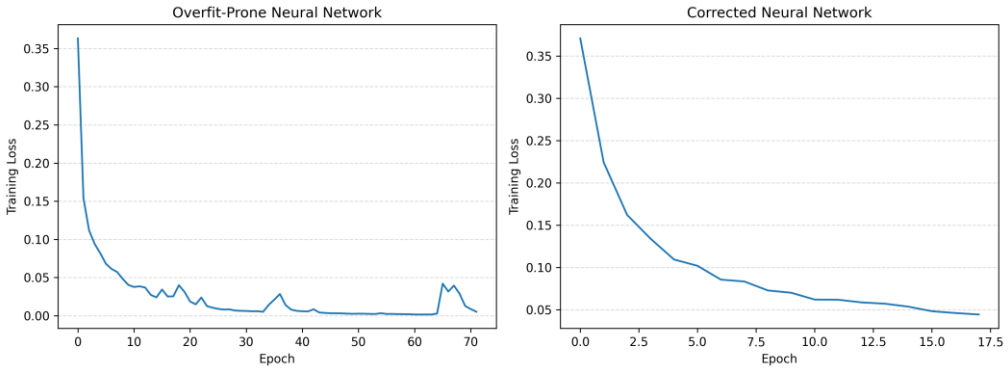


Figure 7 *Neural-network training-loss curves for an overfit-prone model and a corrected model using early stopping and L2 regularization.*

Figure 7 compares the training behavior of a larger neural network, MLP (128, 64, 32), trained without early stopping and with minimal regularization, and the selected MLP (32, 16) trained with L2 regularization ($\alpha = 0.01$) and early stopping. The overfit-prone model trained for 72 iterations, whereas the corrected model stopped after 18 iterations. Early stopping therefore reduced unnecessary training and produced a smoother learning process. However, it did not improve holdout performance in this experiment: the larger model achieved F1 = 0.6028 and PR-AUC = 0.8014, compared with F1 = 0.1830 and PR-AUC = 0.7114 for the corrected model. In this case, the term "corrected" refers to training stability rather than better generalization. The result suggests that the selected configuration was limited more by its capacity than by overfitting, and that larger architectures deserve further study. Even so, both neural networks remained below the Random Forest in F1-score and PR-AUC.

The Random Forest model was selected as the final model for this project. Although the Neural Network achieved higher Recall, the Random Forest provided substantially higher Precision, F1-score, and PR-AUC while generating only 6 false positives. In practical fraud-detection applications, excessive false positives can create significant operational costs because legitimate transactions may require unnecessary review. The Random Forest therefore provided the best balance between fraud detection capability and false-alarm control.

The analysis also indicates that transaction patterns captured by the anonymized principal-component features contain useful information for distinguishing fraudulent and legitimate transactions. Feature-importance analysis identified variables such as V14, V10, V12, and V11 as the most influential predictors in the Random Forest model.

Overall, the results demonstrate that machine learning can successfully identify rare fraudulent transactions despite severe class imbalance and that Random Forest was the most effective approach evaluated in this study.

3.4 Model Application (a.k.a. deployment)

The final Random Forest model could be integrated into a credit-card transaction monitoring system. Each incoming transaction would be evaluated in real time and assigned a fraud probability score. Transactions with a predicted fraud probability above a predefined threshold could be flagged for additional review or temporarily blocked until verification is completed.

The model may help financial institutions identify suspicious transactions before financial losses occur. Because the model achieved Precision = 0.9302 on the holdout set, most transactions it flags are expected to be actual fraud rather than false alarms. The model could therefore reduce the workload of fraud analysts while improving detection efficiency.

The feature-importance analysis suggests that several anonymized transaction characteristics, including V14, V10, V12, and V11, contribute strongly to fraud identification. Although the original meanings of these variables are unavailable because of anonymization, the results indicate that unusual transaction patterns captured by these features are associated with fraudulent activity.

A practical deployment strategy would involve assigning a risk score to every transaction. For example, transactions with predicted fraud probabilities above 0.90 could be automatically blocked, transactions between 0.50 and 0.90 could be sent for manual review, and transactions below 0.50 could be approved automatically. Such an approach would allow organizations to prioritize investigative resources toward the highest-risk transactions.

On the holdout test set, the Random Forest model correctly identified 80 of the 98 fraudulent transactions while generating only 6 false positives. Compared with a trivial classifier that detected no fraud cases, the model provides substantial operational value by identifying suspicious activity while minimizing disruptions to legitimate customers.

Table 7 *Examples of fraud-risk decision thresholds*

Predicted Fraud Probability	Decision
0.05	Approve Automatically
0.35	Approve Automatically
0.72	Manual Review
0.94	Temporarily Block
0.99	Temporarily Block

Table 7 illustrates one possible deployment strategy. Transactions with low predicted fraud probabilities are automatically approved, while medium-risk transactions

are forwarded for manual investigation. High-risk transactions are temporarily blocked pending verification. Such a workflow allows fraud analysts to focus on the most suspicious transactions while minimizing unnecessary review of legitimate customer activity. Compared with the trivial classifier, which detected no fraudulent transactions, the final Random Forest model could substantially reduce undetected fraud cases and improve operational efficiency by directing investigative resources toward the highest-risk transactions. The low false-positive rate would also help reduce unnecessary transaction reviews and customer disruptions.

4 Conclusion

This project investigated credit card fraud detection as a highly imbalanced binary classification problem using a publicly available dataset containing 284,807 transactions and 30 anonymized input features. The CRISP-DM framework was used to guide the project from data understanding through model evaluation. Exploratory analysis confirmed the severe class imbalance, with fraudulent transactions representing only 0.1727% of all observations. Because of this imbalance, model performance was evaluated primarily using Precision, Recall, F1-score, ROC-AUC, and PR-AUC rather than accuracy alone.

Several machine-learning approaches were evaluated, including Logistic Regression, Random Forest, and Multilayer Perceptron neural networks. Class weights were incorporated to help address the imbalance of the fraudulent class. The Random Forest model achieved the strongest overall performance on the holdout dataset, obtaining Precision = 0.9302, Recall = 0.8163, F1-score = 0.8696, ROC-AUC = 0.9803, and PR-AUC = 0.8568. Although the Neural Network achieved higher Recall, it generated substantially more false positives and produced a lower F1-score. Consequently, the Random Forest model was selected as the final model.

The results answer the research question by demonstrating that machine-learning methods can effectively identify fraudulent transactions despite extreme class imbalance. Furthermore, for this dataset, the Random Forest approach outperformed the evaluated neural-network configurations.

Several limitations should be acknowledged. The dataset contains anonymized principal-component features, preventing direct interpretation of the original transaction characteristics. In addition, the data represents historical transactions collected from a single source and may not fully reflect evolving fraud patterns encountered in operational environments.

Future work could investigate additional ensemble-learning methods, advanced neural-network architectures, alternative sampling strategies, and threshold-optimization techniques. Future studies could also evaluate the models using more recent transaction data to assess long-term generalization and robustness against changing fraud behavior.

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